GOH KUN MING

AI Research | Data Science | Machine Learning Engineering | Generative AI

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PROFESSIONAL SUMMARY

Applied AI and Analytics student with experience in machine learning, computer vision and retrieval-augmented generation (RAG). Public Python and R portfolio spans forecasting, clustering, regression and natural language processing (NLP). Combines full-stack AI development with statistical analysis, controlled experimentation, reproducible evaluation and responsible AI practices.

EDUCATION

Diploma in Applied AI and Analytics (DAAA)
Expected 2027

Singapore Polytechnic, School of Computing (SoC)

- GPA: 3.95/4.00, including three consecutive 4.00/4.00 semesters.
- Earned Distinction in 9 modules (top 5%) and A grades in 15 modules. Distinctions include Deep Learning, Statistics for Data Science, Mathematics for AI and DevOps for AI.

PROFESSIONAL CERTIFICATES AND SPECIALISATIONS

- IBM AI Engineering Professional Certificate | Oct 2026
- IBM Deep Learning with PyTorch, Keras and TensorFlow Professional Certificate | Oct 2026
- IBM RAG and Agentic AI Professional Certificate | Oct 2026
- Google Advanced Data Analytics Professional Certificate | Oct 2026
- Google AI Professional Certificate
- Mathematics for Machine Learning Specialization | Imperial College London | Oct 2026
- AI Foundations for Business Professionals Specialization | University of Oxford | Oct 2026
- Building AI Agents and Agentic Workflows Specialization | IBM | Sep 2026

EXPERIENCE

Research Engineer
Oct 2026 - Present

NCS Pte Ltd, Pragnition Labs | Internship

- Conduct applied AI research, review academic literature and translate business and technical requirements into research questions, testable hypotheses and experiments.
- Design, develop and evaluate AI proofs of concept using machine learning, software engineering, data analytics and automation, assessing accuracy, reliability, latency, computational cost, security and maintainability.
- Collaborate with multidisciplinary teams to refine research directions, communicate findings and develop technical documentation and reusable components that support progression towards production-ready AI systems.

AI Research Intern
Apr 2026 - Aug 2026

RSAF Agile Innovation Digital (RAiD), AETHER | Internship

- Reduced false positives by 72% versus a higher mAP alternative, with an offline GPU benchmark of 22.18 FPS. Trained and compared neuromorphic-camera drone detectors on 90,000 training tiles; the selected model achieved 0.760 recall and 0.3049 false positives per frame on an official held-out test set.
- Designed and executed a controlled replay evaluation of 14 RGB, thermal and LiDAR sensor-fusion strategies across 1,442 synchronised frames, 7 sensor subsets and 9 environmental conditions. LiDAR + Thermal Agentic Fusion scored 0.4511 versus 0.4493 for conditional fusion, a 0.0018 gain too small to support a claim of clear superiority.
- Curated approximately 500 GB of geospatial data from 15 sources for downstream modelling, identifying 2,139 image-label pairing issues before producing 13,487 segmentation samples and 56,096 classification records.
- Helped coordinate an 8-person team across AI, hardware, command-and-control and systems integration, and worked with 5 partner institutions spanning universities and polytechnics on sensing and digital-twin workstreams.

AI and Data Solutions Researcher
Mar 2026

Singapore Polytechnic, Department of Communications (COMMS) | Part-time

- Developed a proof-of-concept departmental platform intended for ACER, covering membership, events, attendance, finance and learning journeys across 4 user roles, 11 workflow areas and 32 application routes.
- Designed an explainable 0 to 100 candidate-matching model using 8 operational signals and selection constraints to balance audience fit, reliability, experience and opportunity allocation.
- Created a GPT-4.1 visitor information chatbot for 4 audience groups, grounded in 14 documents, 1 institutional website and 9 curated Q&A entries. Evaluated it with 30 students and additional staff.
- Designed a Supabase backend with 39 relational tables and 175 row-level security policies, then verified core workflows through 238 automated tests, 31 browser scenarios and 4 CI/CD workflows.

EXPERIENCE

Student Developer
Sep 2025 - Mar 2026

Singapore Polytechnic, School of Computing (SoC) | Project-based

- Built and evaluated an institutional retrieval-augmented generation (RAG) assistant using Claude 3 Haiku and Cohere embeddings through Amazon Bedrock, pgvector, BM25, reciprocal rank fusion and large language model (LLM) reranking. Responses used a median of 4 sources and 8 contextual records.
- Analysed 4,056 logged responses. Overall, 93.6% returned without a clarification, knowledge gap or safety fallback. Among 519 RAG responses, 94.8% avoided a knowledge gap fallback, with no empty answers.
- Delivered the full-stack application through 53 FastAPI endpoints and 7 Next.js pages, integrating Amazon S3 and Textract ingestion, encrypted data pipelines, persistent conversational memory and 183 structured institutional records.
- Validated retrieval, authentication and safety-critical workflows with 312 pytest definitions and 11 browser scenarios, supported a 3-day pilot with 287 responses across 72 conversations, and produced a 30-page handover guide for Work Plan Seminar 2026.

ENTREPRENEURIAL EXPERIENCE

Data Governance & Technical Advisor
May 2026 - Present

Wagglo SG | Volunteer

- Advise on privacy governance and responsible data stewardship for Wagglo, an AI-powered dog compatibility and social matching platform.

Co-Founder, Data & Intelligent Systems
Feb 2026 - Mar 2026

Wagglo SG | Self-employed

- Co-founded and launched Wagglo on the App Store and Google Play. Wagglo was later featured in The Straits Times.
- Analysed 104 survey responses using weighting, resampling and bootstrap estimation to correct a 3:1 age imbalance, identifying compatibility (73.5%) and safety (72.5%) as users' leading concerns.
- Designed an interpretable matching and recommendation system using 9 preference dimensions, Top-K retrieval and evidence-gated personalisation across 8 behaviour types.

Founder and Project Lead
Mar 2025 - Mar 2026

Gentle Care Messenger (GCM) | Self-employed

- Led a 4-person team and presented GCM at the NUS HumanITy Challenge, defining a non-diagnostic system aligned with UN SDG 3.4, mandatory human escalation and no autonomous clinical intervention.
- Created a transparent risk engine using 18 distress phrases, 6 contextual indicators and deterministic 0 to 100 scoring across 4 risk categories, paired with human review across 4 priority levels and 5 review states.
- Architected a research prototype spanning 4 application surfaces, 6 shared packages, 12 API routes and 15 relational models, with role-based consent, review and audit workflows supported by 27 security and operational documents and 10 passing tests.

AWARDS AND LEADERSHIP

- Awards: National Youth Achievement Award (NYAA) Gold Award, 2026; Singapore Polytechnic Director's Honour Roll (AY2025/2026); Boys' Brigade Founder's Award and SP Chua Medal (Best Boy), 2023.
- Cadet Lieutenant, The Boys' Brigade in Singapore (2024 - 2026) | Volunteer: mentored 30+ youths aged 13 to 17, taught Python fundamentals and supported camps, leadership lessons and character programmes.

TECHNICAL SKILLS

Programming	Python, R, SQL, JavaScript and TypeScript
Research Areas	Generative AI, Agentic AI, Large Language Models, Computer Vision, Perception Systems, Quantum Machine Learning and Responsible AI
Research Methods	Experimental Design, Model Evaluation, Error Analysis, Reproducible Research and Technical Documentation
Data Science Methods	Exploratory Data Analysis, Feature Engineering, Regression, Classification, Clustering, Time Series Forecasting and Statistical Analysis
ML and AI Libraries	PyTorch, TensorFlow/Keras, scikit-learn, Qiskit and OpenCV
Generative AI	Retrieval-Augmented Generation (RAG), Embeddings, Vector Search, LangChain, Amazon Bedrock and OpenAI API
AI Coursework	Transformer Models, Model Fine-Tuning and Multi-Agent Workflows
Data and Applications	Pandas, NumPy, FastAPI, React, Next.js, PostgreSQL, Supabase, Plotly, Power BI and Tableau
Cloud and Engineering	AWS, Azure, Docker, Terraform, Git/GitHub, CI/CD, pytest, Playwright, Linux and REST APIs